

Machine Learning 2

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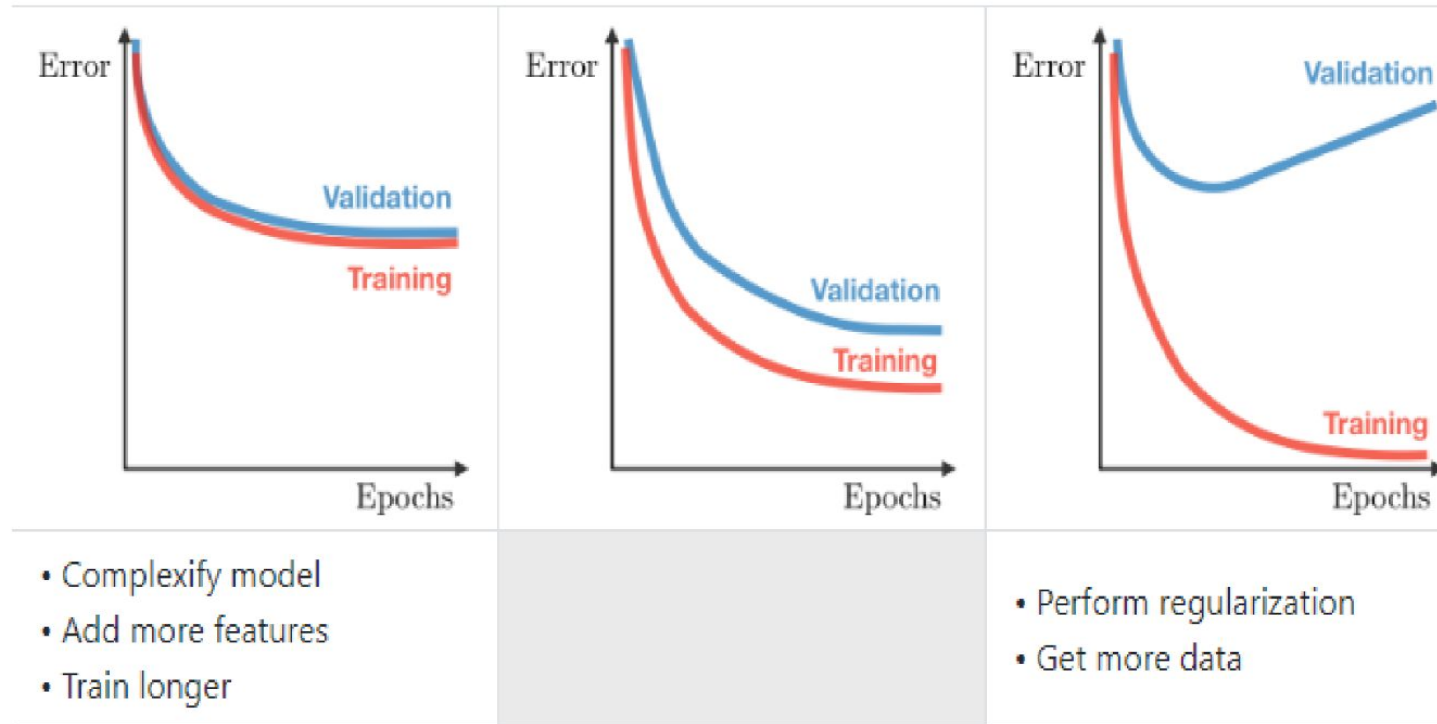
Learning Theory

What is today's lecture about?

- Overfitting and Underfitting
- Bias-variance decomposition and tradeoff
- Model Selection and Cross-Validation
- Regularization

Overfitting and Underfitting

Overfitting and Underfitting



How do we detect undefitting?

- Model's quality is too low
- Loss does not converge
- Val loss might be lower

Causes and Remedy for Underfitting

- Model's too simple → complexify your model!
- Learning time is too inadequate → train longer!
- Too much of regularization → start simple, complexify the pipeline step by step
- Inadequate features → run some tests, add more features

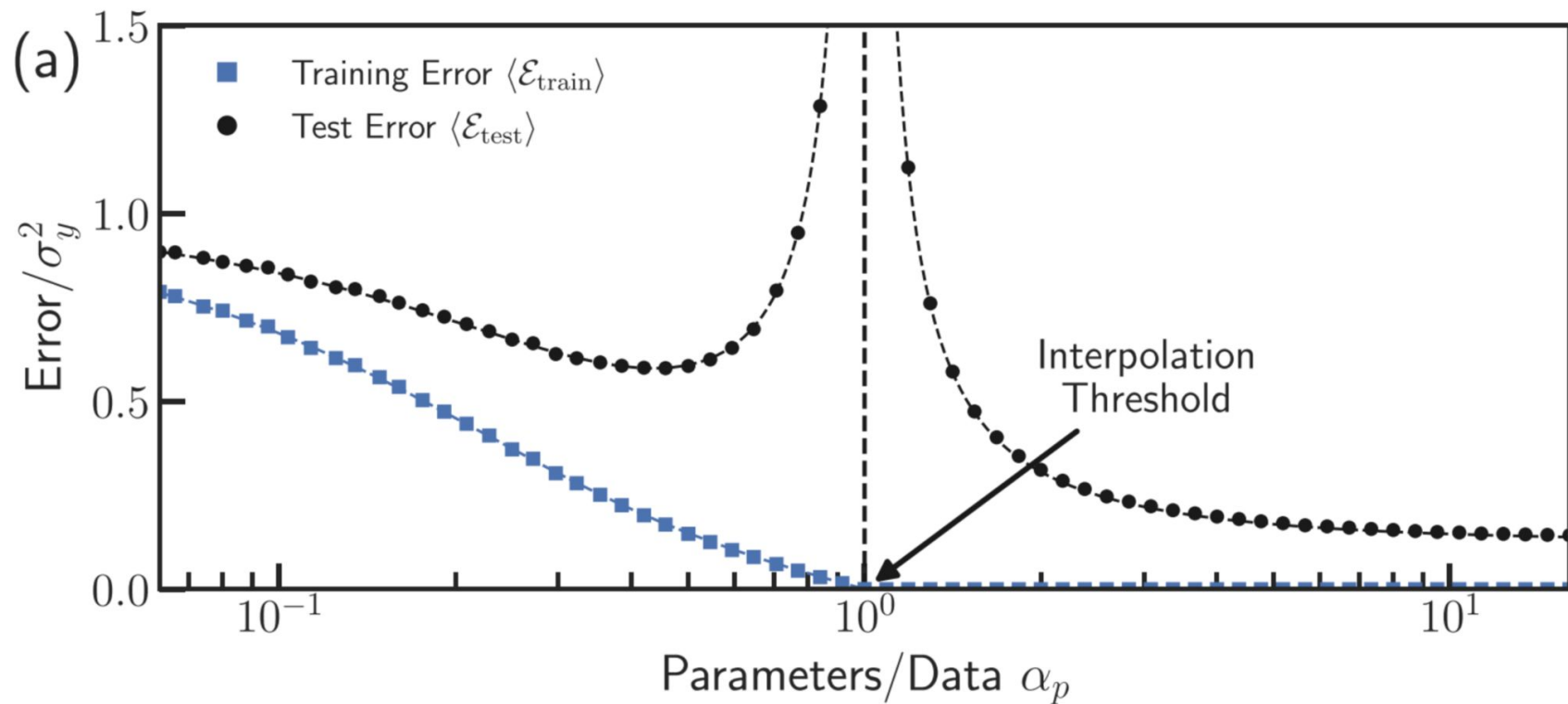
How do we detect overfitting?

- Very high training accuracy, significantly lower validation accuracy
- Unstable accuracy on different test subsets

Causes and Remedy for Overfitting

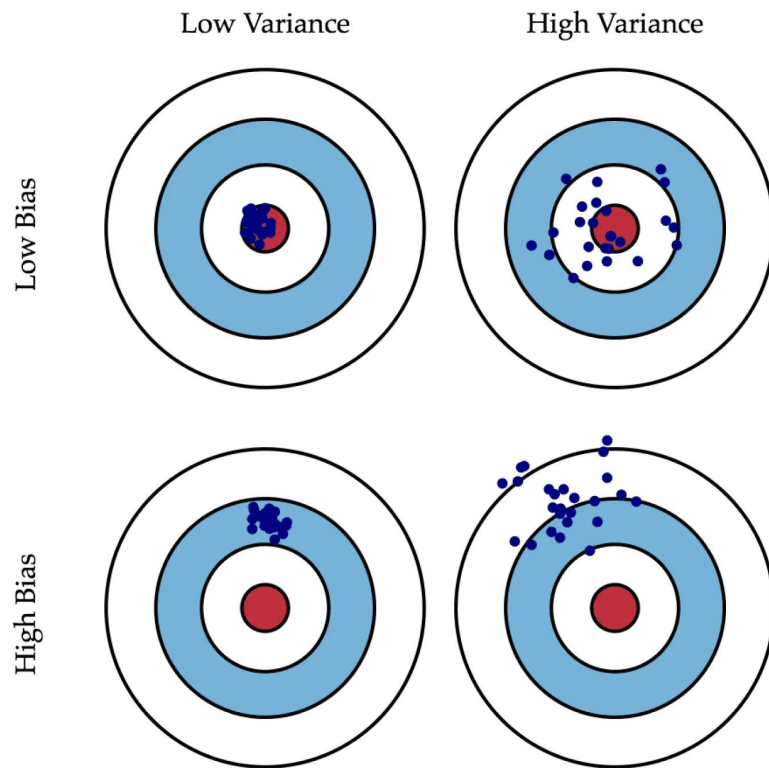
- Model is too complex → make it smaller
- Model it too complex → add regularizations
- Too little of a dataset → add more data/augmentations
- Too much of a training iterations → add early stopping

Extreme case



Bias-Variance Decomposition

Bias and Variance



Bias and Variance

- **Error due to Bias:** The error due to bias is taken as the difference between the expected (or average) prediction of our model and the correct value which we are trying to predict
- **Error due to Variance:** The error due to variance is taken as the variability of a model prediction for a given data point

Bias and Variance

- $y = f(x) + \varepsilon$, where f is a general function, and ε is a random noise
- $\mathbb{E}\varepsilon = 0$, $\text{Var}\varepsilon = \mathbb{E}\varepsilon^2 = \sigma^2$.

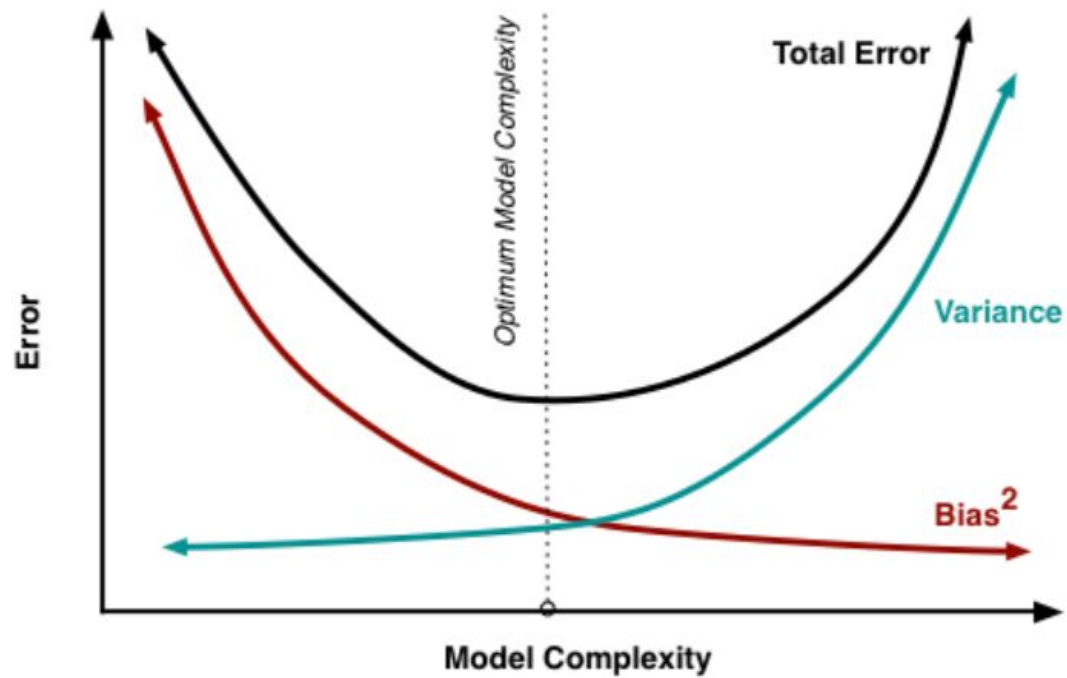
$$Q(a) = \mathbb{E}_x \mathbb{E}_{X, \varepsilon} [y(x, \varepsilon) - a(x, X)]^2$$

Bias and Variance

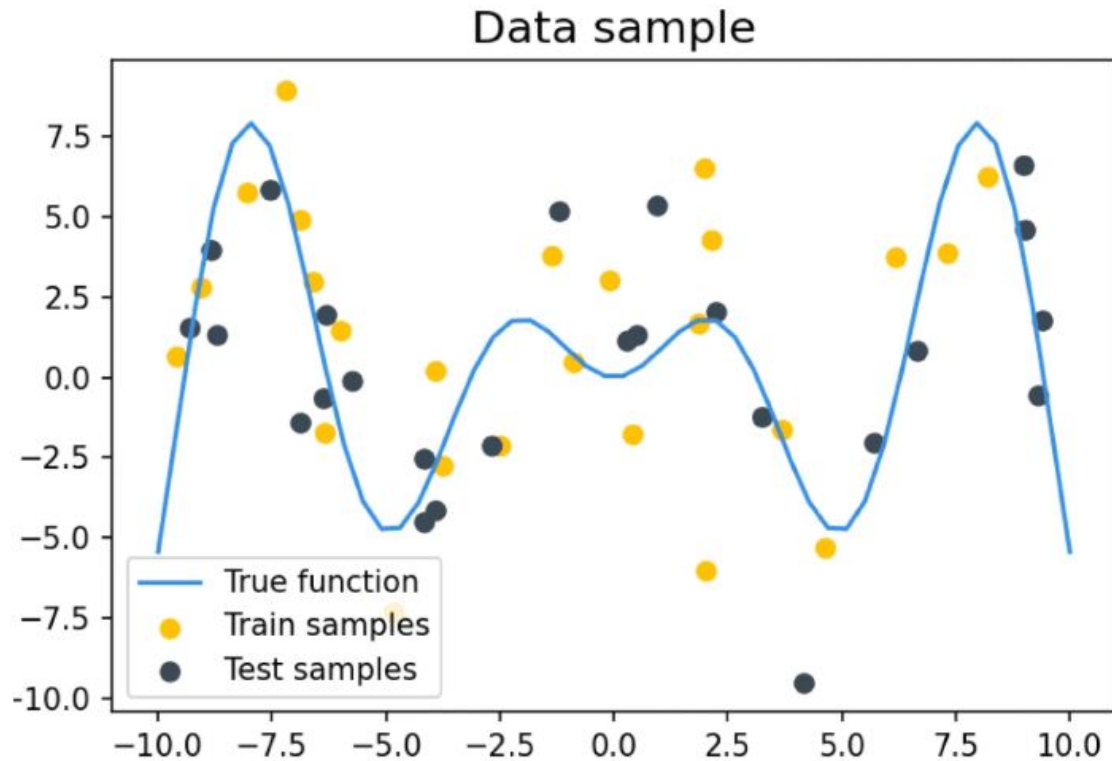
$$Q(a) = \mathbb{E}_x \mathbb{E}_{X, \varepsilon} [y(x, \varepsilon) - a(x, X)]^2$$

$$= \mathbb{E}_x \text{bias}_X^2 a(x, X) + \mathbb{E}_x \text{Var}_X [a(x, X)] + \sigma^2$$

Bias and Variance

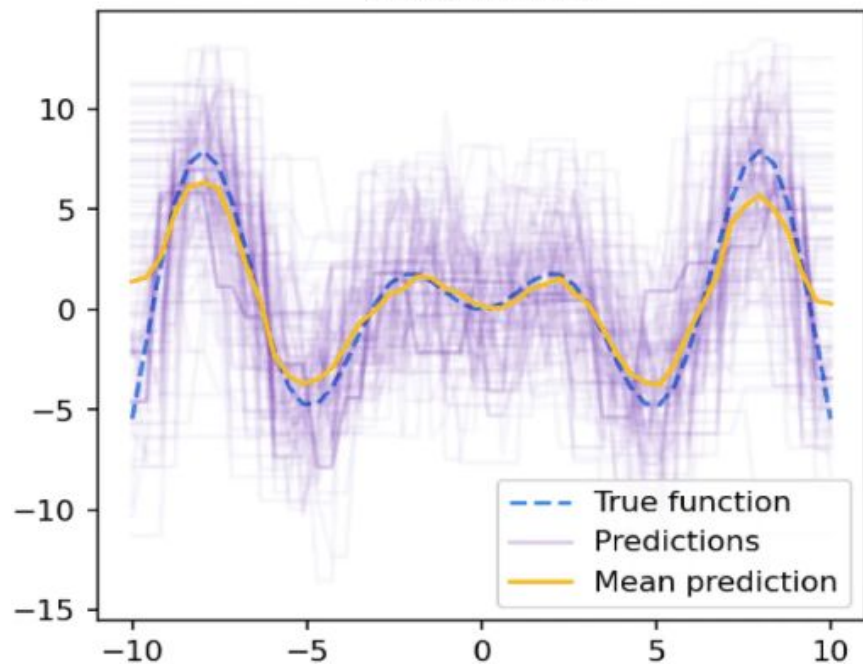


How to reduce bias and variance?

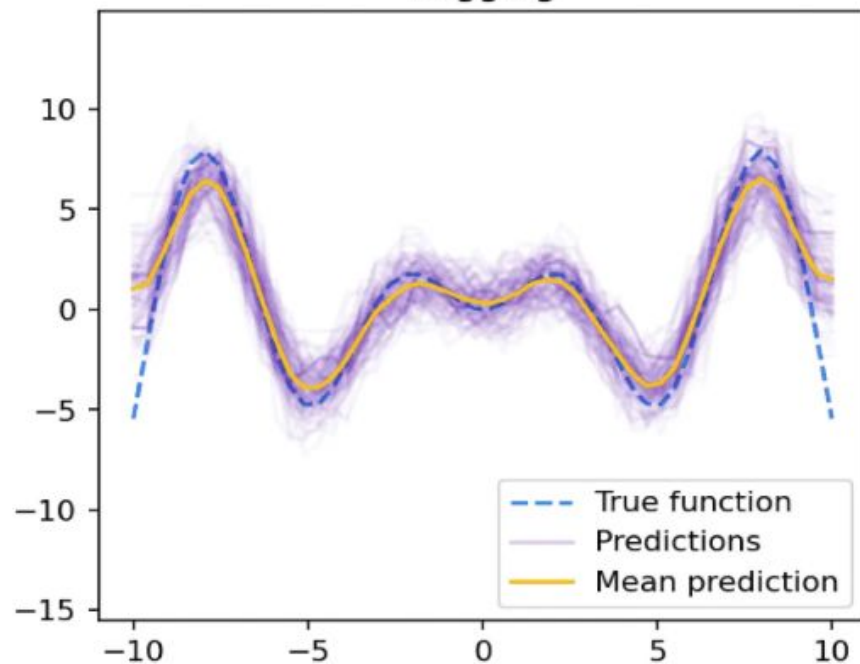


How to reduce bias and variance?

Decision tree



Bagging

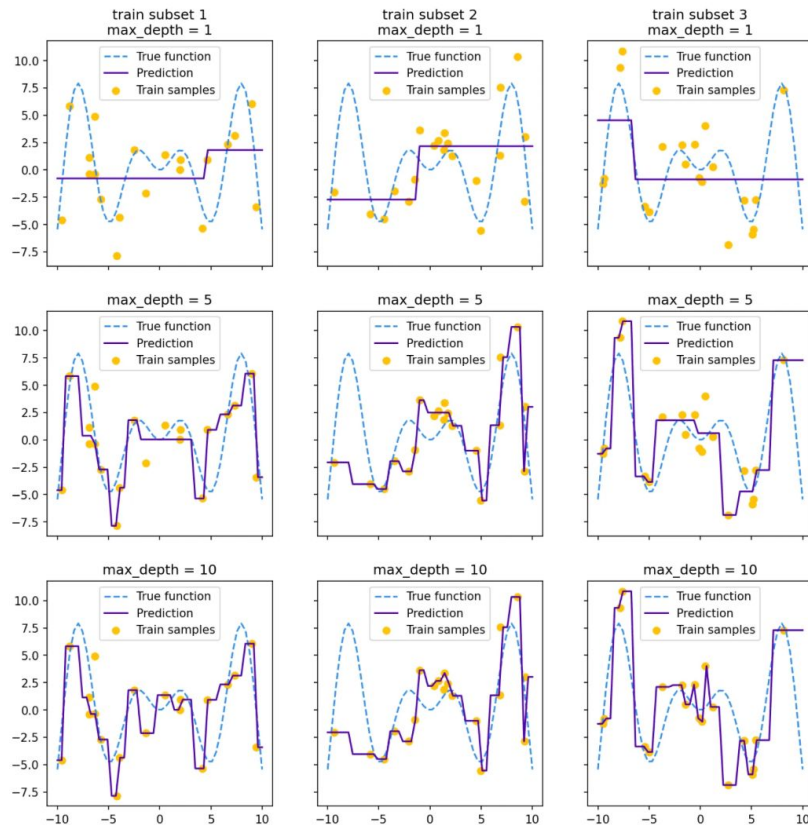


Bagging

- Bagging — **B**ootstrap **A**ggregating
- Bias of the bagging model (RF, for example) is equal to the bias of each individual model of the composition
- Variance of the bagging model is **k** times lower than the variance of the each (out of k) individual model of the composition

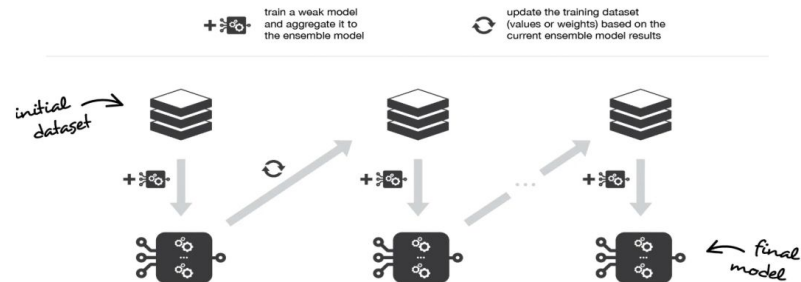
Random Forest

- Shallow Decision Trees has high bias
- The deeper the Tree, the lower the bias and higher the variance
- But the bagging allows us to reduce the variance
- Use **deep** trees in Random Forest
- **Reduce variance** by **increasing** the number of estimators (trees)



Boosting

- Reduce the error of the previous model
- Use **weak** estimators (e.g. shallow trees)
- **Increase** the amount of estimators
- Be careful! More estimators = higher variance



Stacking

Model Selection

Model Selection

- Decide which model is better on test set
- But how to determine whether those $x\%$ difference in a big deal?

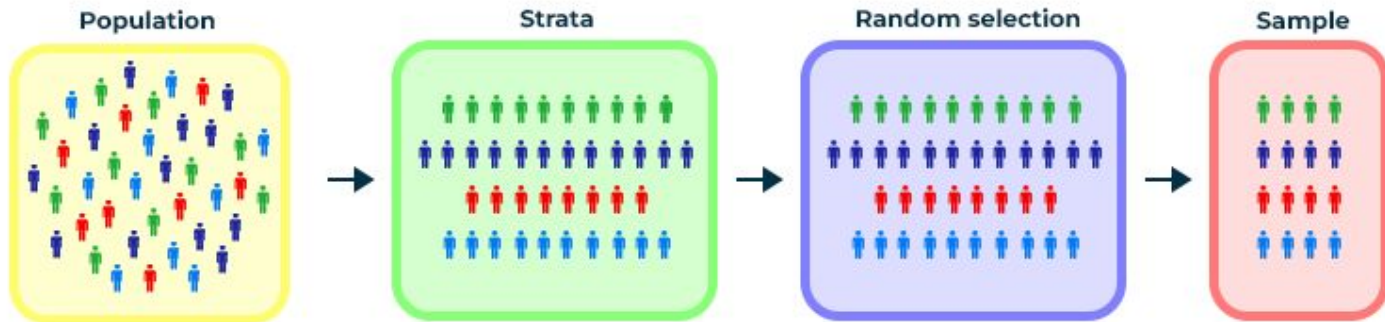
Data Splitting

We usually split the data into train and test. It is also worth noticing that an additional validation split is viable

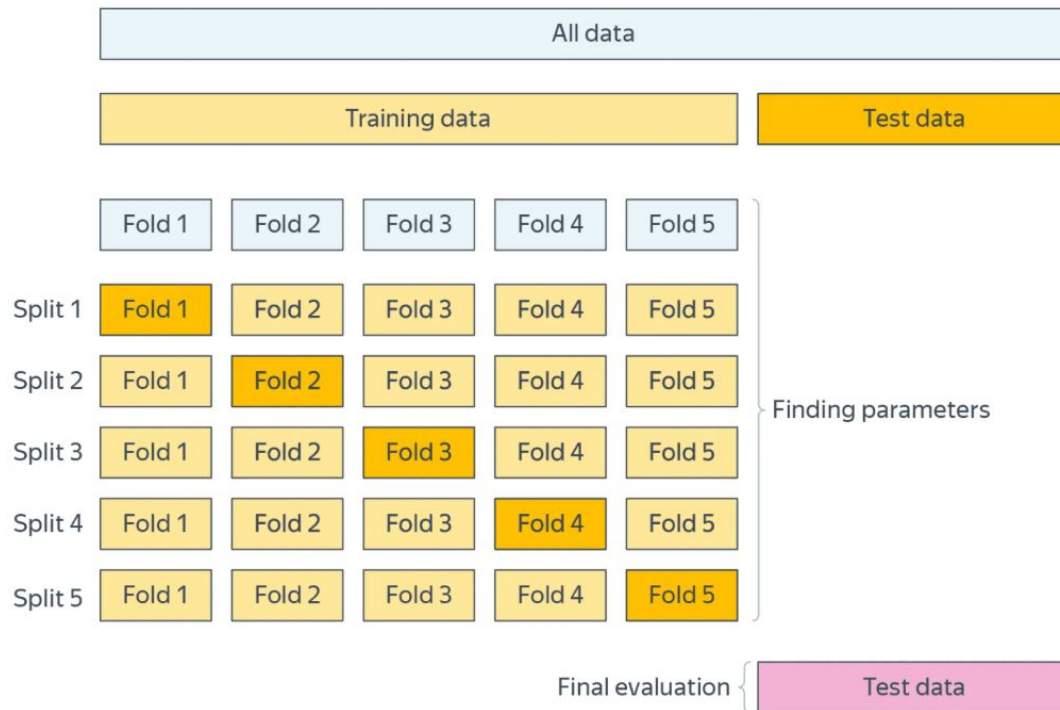


Stratification

Stratified sampling



k-Fold Validation



k-Fold Validation

- Use different models
- Train each model on different splits
- Measure performance across all folds
- Compute the mean and std of the metrics
- Now you have more **robust** estimation of model's performance

Regularization

Regularization

Regularization is a process that converts the answer to a problem to a simpler one. It is often used in solving ill-posed problems or to prevent overfitting.

We often discuss **explicit** regularization leaving **implicit** one behind

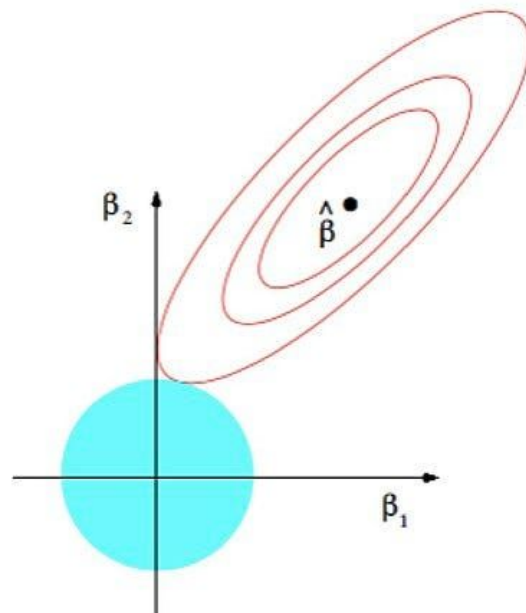
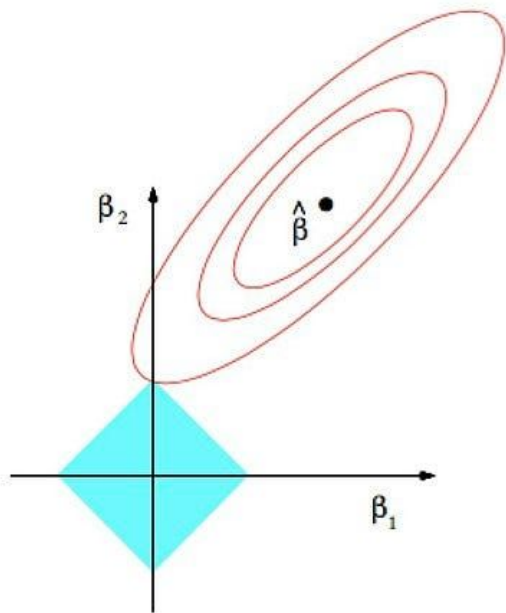
Explicit Regularization

- Constraints (spectral normalization/gradient clipping)
- Penalties (l_1/l_2 norms)

Implicit Regularization

Implicit regularization is all other forms of regularization. This includes, for example, early stopping, using a robust loss function, and discarding outliers. Implicit regularization is essentially ubiquitous in modern machine learning approaches, including stochastic gradient descent for training deep neural networks, and ensemble methods

L1 and L2 regularization



L1 and L2 regularization

$$\min_w L(f, X, y) = \min_w (\|Xw - y\|_2^2 + \lambda \|w\|_k^k)$$

$$\|w\|_1^1 = |w_1| + \dots + |w_D|$$

$$\|w\|_2^2 = w_1^2 + \dots + w_D^2$$

Next week

- More on regularization
- ML experiment pipelines and how to design them